

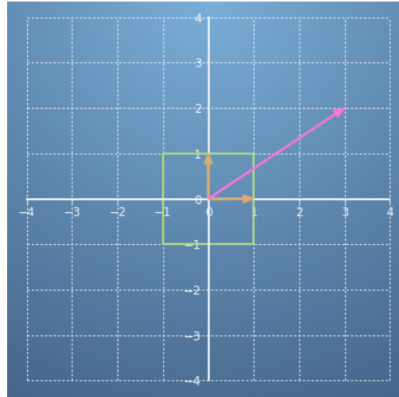
1. Matrices make transformations on vectors, potentially changing their magnitude and direction.

1 point

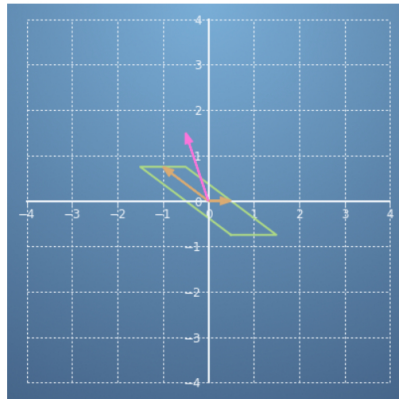
If we have two unit vectors (in orange) and another vector,

$$\mathbf{r} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \text{ (in pink),}$$

before any transformations - these look like this:



Take the matrix, $A = \begin{bmatrix} 1/2 & -1 \\ 0 & 3/4 \end{bmatrix}$, see how it transforms the unit vectors and the vector, $\mathbf{r}$,



What new vector, $\mathbf{r}'$, does A transform $\mathbf{r}$ to? Specifically, what does the following equal?

$$A\mathbf{r} = \begin{bmatrix} 1/2 & -1 \\ 0 & 3/4 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} =$$

- ☐ $\begin{bmatrix} -1/2 \\ 3/2 \end{bmatrix}$
- ☐ $\begin{bmatrix} 3/2 \\ -1/2 \end{bmatrix}$
- ☐ $\begin{bmatrix} 3/2 \\ -3/4 \end{bmatrix}$
- ☐ $\begin{bmatrix} -3/2 \\ 3/2 \end{bmatrix}$

2. Let's use the same matrix, $A = \begin{bmatrix} 1/2 & -1 \\ 0 & 3/4 \end{bmatrix}$, from the previous question.

1 point

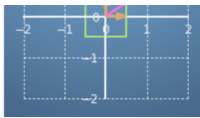
Type an expression for the vector, $\mathbf{s} = A \begin{bmatrix} -2 \\ 4 \end{bmatrix}$.

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1 # Replace a and b with the correct values below:
2 s = [-5 , 3]
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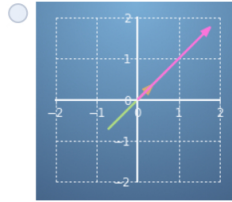
3. Select the transformation which best corresponds to the matrix, $M = \begin{bmatrix} -1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$.

1 point

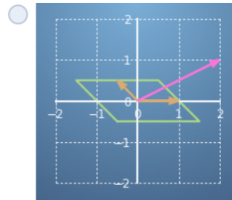




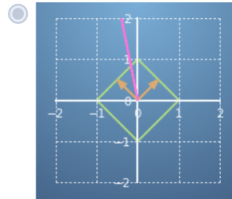
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4. A digital image can be stored by putting lots of coloured pixels at their particular coordinates on a grid.

1 point

If we apply a matrix transformation to the coordinates of each of the pixels in an image, we transform the image as a whole.

Given a starting image (such as this one of "The Ambassadors" [1533] by Hans Holbein the Younger),



which is made up of 400×400 pixels, if we apply the same transformation to each of those 160,000 pixels, the transformed image becomes:





Pick a matrix that could correspond to the transformation.

- ☒ $\begin{bmatrix} \sqrt{3}/2 & -1/2 \\ 1/2 & \sqrt{3}/2 \end{bmatrix}$
- ☐ $\begin{bmatrix} \sqrt{3}/2 & \sqrt{3}/2 \\ 1/2 & 1/2 \end{bmatrix}$
- ☐ $\begin{bmatrix} -1/2 & 0 \\ 0 & \sqrt{3}/2 \end{bmatrix}$
- ☐ $\begin{bmatrix} 1/2 & 0 \\ -\sqrt{3}/2 & 1/2 \end{bmatrix}$

5. At the bottom of the “The Ambassadors”, in the middle of the floor, there is a skull that Holbein has already applied a matrix transformation to!

1 point

To undo the transformation, build a matrix which is firstly a shear in the y direction followed by a scaling in y direction. i.e., multiply the matrices,

$$M = \begin{bmatrix} 1 & 0 \\ 0 & 8 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1/2 & 1 \end{bmatrix}$$

```
1 # Replace a, b, c and d with the correct values below:
2 M = [[1, 0],
3      [-4, 8]]
```

6. Use your answer from the previous question to transform the skull back to normal. Change the values of the matrix and press Go! to score on this question.

1 point

You can also use this example to experiment with other matrix transformations. Try some of the ones in this quiz. Have a play!

$M = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Go!

Reset



[Expand](#)